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Review article

Artificial intelligence and its performance impacts in the oil and gas industry: Challenges, insights, and evaluation approaches

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ABSTRACT

Oil and gas (O&G) companies are experimenting with artificial intelligence (AI) tools in exploration, production, and trading, but few are able to provide long-term benefits. This thematic review distinguishes basic AI adoption from AI realization—the sustainable ability to embed, scale, and profit from AI through coherent data, skills, and routines. Three research strands are synthesised. First, the literature consistently identifies three driver groups—people, technical, and managerial—that accelerate or hinder realization. Second, progress is usually gauged with three measurement families: text mining of corporate filings, proxy indicators such as AI-related patents or robot density, and survey-based capability indices. Third, realised AI affects three performance domains: operational, financial, and environmental. Within O&G, value chain stage (upstream, midstream, downstream) and ownership model (state-owned and privately owned) modulate both measurement validity and outcome strength. These strands are integrated into a conceptual synthesis that links drivers to gauges and gauges to results, stressing triangulation across indicators to limit bias. Triangulated dashboards linking capability scores to these outcomes expose trade-offs, such as efficiency gains that raise energy use. Recommended actions for practitioners include: connecting operational, financial, and sustainability databases; deploying an annual "AI readiness" survey covering data, talent, and governance; and adopting shared reporting rules that enable cross-company learning. For scholars, the paper highlights measurement gaps, suggests triangulated research designs, and maps boundary conditions needing further study. Together, the insights give executives a firmer basis for allocating AI budgets and give researchers a clearer roadmap toward comparative, evidence-based evaluation of AI value in complex, asset-intensive settings.

Contents

1. Introduction . . . 2
2. Conceptual foundations . . . 3
   2.1. AI-related definitions . . . 3
   2.2. O&G industry differences . . . 3
   2.3. AI applications in O&G industry . . . 4
3. Review scope and methodology . . . 4
4. Drivers to AI realization in the O&G industry . . . 5
   4.1. People . . . 5
   4.2. Technical . . . 6
   4.3. Managerial . . . 6
5. Measuring AI realization . . . 6
   5.1. Textual analytics . . . 6
   5.2. Proxy variables . . . 6
   5.3. Empirial capability indices . . . 6
      5.3.1. Resource driven capability . . . 6
      5.3.2. Business requirement capabilities . . . 7

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5.3.3. AI competency . . . 7
5.4. Drivers and AI realization measures . . . 7
5.5. Triangulation rule for AI realization measures . . . 7
6. Performance outcome domains . . . 8
6.1. Triangulation rule for performance measures . . . 8
7. Conceptual synthesis . . . 8
7.1. Recommendations for O&G companies . . . 8
7.2. Recommendations for the O&G industry . . . 9
8. Conclusion and future research . . . 9
CRediT authorship contribution statement . . . 10
Declaration of generative AI and AI-assisted technologies in the writing process . . . 10
Declaration of competing interest . . . 10
Data availability . . . 10
References . . . 10

1. Introduction

The oil and gas (O&G) industry underpins global energy and industrial activity. Yet it now faces four pressures: volatile prices, rising efficiency demands, stricter decarbonization rules, and heightened stakeholder scrutiny [1,2]. The oil and gas (O&G) industry underpins energy supply and industry worldwide, yet the sector now faces four significant pressures: volatile prices, rising efficiency demands, strict decarbonization rules, and heightened stakeholder scrutiny. Many firms view advanced digital tools, especially Artificial Intelligence (AI), as a means to reduce costs, enhance reliability, and lower emissions across the various stages of the O&G value chain [3,4]. According to a survey study, 70% of AI projects in general industries fail or have little effect [5]. Despite this, major O&G industry producers have funded AI startups and established data-science teams, but it is still difficult to measure and find clear, long-lasting benefits [6,7].

This creates a 'technology-optimism' puzzle: firms spend heavily on digital tools, yet productivity gains remain slow or unclear [8,9]. The problem is exacerbated by a 'technology optimism puzzle,' in which high digital expenditures coexist with sluggish or ambiguous productivity development. Executives ask: Do these AI projects really pay off? Answering needs two things. First, sound ways to gauge how far a firm has moved from simple AI adoption to full AI realization—that is, the ability to embed, scale, and profit from AI using data, skills, and routines [10]. Mere adoption is a first step; however, AI realization signals depth and clear impacts.

Researchers still lack consensus on how to measure AI realization and its pay-offs—an even harder task in O&G. As Fig. 1 shows, performance depends on how deeply AI is realized, so measuring realization is essential to assess impact. Researchers have not reached common ground on how to gauge AI realization or its business pay-offs, and the task is tougher still in O&G, a sector with distinctive characteristics. Fig. 1 illustrates that organisational performance is shaped by the depth to which AI is realised; therefore, gauging that realization is essential for clarifying its true impact on results.

Thus, this paper asks four questions:

1. **RQ1:** Which organisational and contextual factors help or hinder the move from initial AI use to sustained AI realization?
2. **RQ2:** How to gauge AI realization in an organization?
3. **RQ3:** How to assess AI realization impacts on a company's performance?
4. **RQ4:** Which features of the O&G sector make it easier or harder to gauge AI realization and understand its effects?

To address these questions, a broad, but not fully systematic, review of the available literature was conducted. Three strands that are usually treated separately are integrated: (1) drivers of successful AI realization, (2) methods to gauge realization, and (3) performance outcomes once

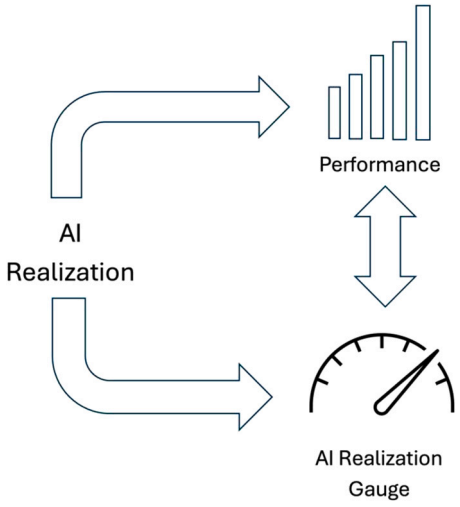


Fig. 1. AI Realization vs. Performance Outcomes.

AI is realized. Insights from the nexus of these three topics are mapped onto the specific context of the O&G industry. Accordingly, this article presents a thematic literature review, organizing cross-industry evidence on AI realization and its performance effects into three themes—drivers, measures, and outcomes—and then contextualizes these themes within the O&G sector.

The importance of this study stems from the scarcity of publications in this area. Only one empirical O&G study has related measured AI use to overall firm performance, despite numerous case reports, such as fault detection in seismic data, drone-based pipeline checks, and AI-driven trading desks [11,12]. Its scope is limited [13] because it only surveys eleven Nigerian firms and overlooks the O&G industry-specificities, such as the value chain stage and ownership type.

For researchers, the paper offers a structured starting point to design empirical studies on measuring AI realization and assessing its impacts in O&G firms. For industry decision makers, it underscores that there is no "one-size-fits-all" recipe: valid measures and credible returns depend on company context, data realities, and strategic choices. By showing how drivers, measures, and outcomes fit together, the review helps executives prioritise investments, set realistic expectations, and justify AI spending with evidence rather than hype. Finally, this study provides recommendations to O&G professional groups overseeing this business to accelerate the successful adoption of AI in this sector.

The paper proceeds as follows. Section 2 establishes the conceptual foundations by defining key terms and highlighting the unique context of the O&G industry. Section 3 outlines the review methodology adopted in this study. Section 4 then examines the drivers of AI realization (people, technical, and managerial); Section 5 reviews methods for

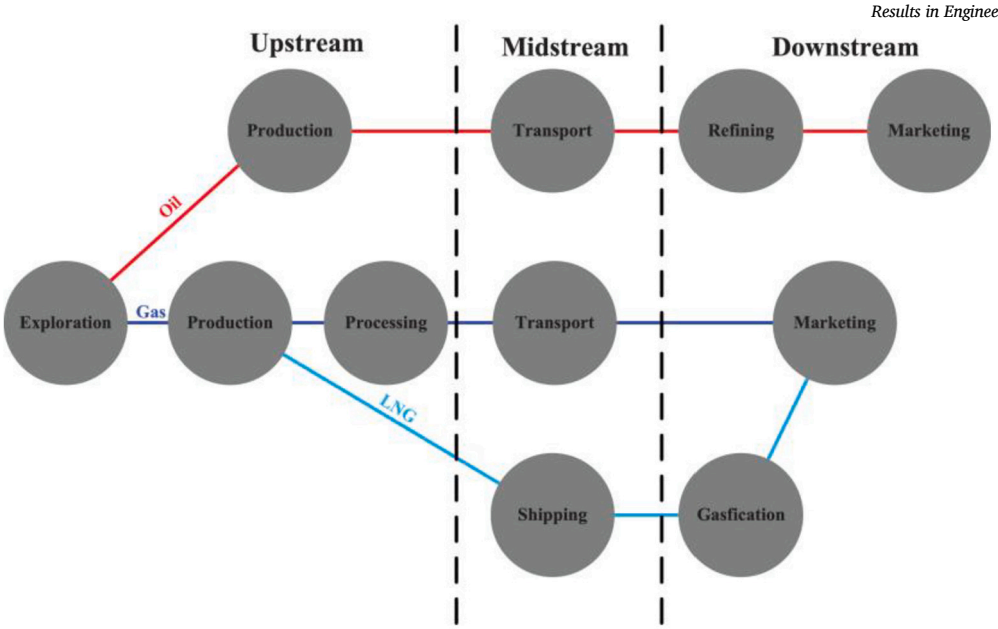


Fig. 2. O&G Industry value Chain, adapted from [25].

measuring AI realization (textual analytics, proxy variables, and capability indices); and Section 6 details the performance outcome domains (operational, financial, and environmental). Section 7 presents a conceptual synthesis integrating these themes into a framework specific to the O&G sector. Finally, Section 8 concludes with implications and a future research agenda.

2. Conceptual foundations

This section lays the theoretical groundwork on which the rest of the paper is built. Firstly, it provides a straightforward definition of AI that is suitable for firm-performance work. Second, it lists the structural features that make the O&G industry different from other industries. Lastly, several AI applications in the O&G industry are presented.

2.1. AI-related definitions

AI is often described as giving computers human-like skills—comprehension, reasoning, problem-solving—so they can handle tasks that normally demand human intelligence [14,15]. It acts as an intelligent agent, interpreting information and taking action much as people learn and decide [16]. Some writers frame AI as a tool for jobs too time-consuming or difficult for people, while others treat it as a broad scientific field or as a practical system capability [17,18]. This study adopts the latter, pragmatic view: AI is an applied discipline that lets computers detect, analyse, infer, and learn from data to meet business and societal goals—matching our focus on measurable firm performance.

Most performance gains flow from machine learning and deep learning techniques [19,20]. Machine learning trains models to spot patterns, draw conclusions, and make predictions via statistical methods [21]. Deep learning goes further, using multi-layer neural networks that mimic brain neurons to uncover complex structures in data [22,23]. In practice, firms combine these methods to build adaptive AI solutions [24].

2.2. O&G industry differences

Segment diversity in the O&G value chain is a main characteristic of this industry, as shown in Fig. 2. Fig. 2 shows three streams: Upstream, midstream, and downstream, and each stream fulfils specific functions and faces peculiar issues and challenges.

The upstream segment, also known as exploration and production, is concerned with identifying, accessing, and extracting crude O&G from under the Earth's surface [26,27]. It includes geological surveys, seismic analyses, and drilling activities. Upstream enterprises are typically assessed according to their reserves, which dictate their prospective production capacities [28]. Main characteristics of the upstream segment include [25]:

1. Necessitates substantial financial investment.
2. Requires specialized skills due to its complexity and technological requirements.
3. High-risk of uncertainty in identifying resources and volatile global oil prices.

The midstream sector connects upstream output with downstream processing. It encompasses the conveyance, storage, and wholesale distribution of O&G [28,29]. Pipelines, shipping, and storage facilities are integral to midstream activities. This sector ensures the transportation of O&G generated by upstream operations to refineries and processing facilities. The main segment characteristics include [28,29]:

1. Regulatory scrutiny is very rigorous.
2. Efficiency of transportation is very important.
3. Environmental and safety hazards linked to transportation.

The last segment is the downstream segment that faces customers. Downstream revolves around the processing and refining of O&G into finished products for consumer utilization, including gasoline, diesel, heating oil, jet fuel, lubricants, and petrochemicals [28,30]. This segment includes refining operations, product distribution, and sales. The downstream segment main characteristics include:

1. Significantly influenced by market demand, consumer behaviour, and price systems.
2. Profitability is intricately linked to the efficacy of refining operations.
3. Success linked to satisfying product demands for both energy and non-energy uses.

In addition to the main differences in the value chain segments, the O&G industry also has several differences regarding the following:

Table 1
Illustrative AI applications across the O&G value chain.

Stage	Application	Reference
Upstream	Prediction of reservoir petrophysical properties	[36]
	Seismic reservoir characterisation	[37]
	Predictive maintenance of pumps	[38]
	Hydraulic-fracturing control	[39]
Midstream	Pipeline leak detection	[40]
	Improved control of transport-facility operation	[41]
	Optimised maintenance in an oil-storage facility	[42]
	Leak size & location detection in blended-gas transmission pipelines	[43]
Downstream	Improving oil blending in a refinery	[44]
	Optimising distribution control in an oil refinery	[45]
	Forecasting crude-oil demand	[46]
	Predicting oil-price changes	[47]

Table 2
Selected AI deployments reported by leading O&G companies.

Company	AI application	Reported impact outcome
TotalEnergies®	Seismic interpretation and modelling (Sismage-CIG)	Faster 3-D reservoir modelling, higher seismic resolution (HPC); method also applied to offshore-wind siting
	Predictive maintenance	Lower rotating-equipment downtime across upstream assets
BP®	Cognitive AI for operations	A decision support system and automation across exploration, refining and marketing
	In-house "AI Factory"	Tighter loop between data science and domain experts; faster time-to-value on drilling and reservoir studies
ExxonMobil®	Predictive maintenance and well development (IBM partnership)	66% reduction in sample-processing time; US$9.6k annual labour savings per site; two months faster well-design cycle.
	Asset-reliability analytics	40% cut in data-preparation effort; potential US$38M saved via lower downtime.

1. Ownership form. National oil companies (NOCs) face policy mandates, bureaucratic processes, and disclosure constraints that can slow or reshape AI roll-out; international oil companies (IOCs) typically have more flexibility and stronger incentives to publicise innovation [31].
2. Capital intensity and risk. Projects in O&G firms are large, multi-year, and tightly regulated; failures carry heavy financial and environmental costs [32,25]. Firms demand clear payoffs and robust governance before scaling AI.
3. Data complexity. Heterogeneous sensors, multiphase flow, and patchy labelling make data preparation a dominant bottleneck [33,34].

2.3. AI applications in O&G industry

Several review papers have looked at different AI applications in the O&G industry. Readers are referred to the following papers for more comprehensive listings of such applications: [1], [35], [6], and [2]. Table 1, however, presents some of these applications divided across the different stages of the O&G value chain.

Beyond the scholarly literature, leading operators—TotalEnergies®, BP®, and ExxonMobil®—also report, via press releases and case studies, how they apply AI. TotalEnergies® has deployed advanced AI models for seismic imaging and predictive maintenance, improving 3-D subsurface models and reducing equipment downtime [48]. BP®'s partnership with AI companies such as Beyond Limits® demonstrates the application of machine learning to enhance reservoir simulations and improve drilling efficiency [49]. ExxonMobil® uses AI to analyse laboratory data from numerous wells and surface facilities, cutting unplanned downtime and boosting asset reliability [50]. Table 2 summarises the reported benefits, illustrating the sector's shift toward intelligent data processing and automation in exploration, drilling and production.

These industry-level advances echo the first calls from international bodies to adopt AI in petroleum engineering during the 1970s [51] and align with the Society of Petroleum Engineers' formal endorsement of AI initiatives in 2009 [51].

Looking ahead, the next wave of AI in O&G is likely to be shaped by multimodal and generative approaches. Multimodal AI integrates diverse data sources—such as seismic images, geological maps, drilling logs, and production records—allowing insights that would be missed if each source were analyzed in isolation. For instance, combining geological property maps with production data through a multimodal machine learning framework has been shown to improve shale gas forecasting [52]. Generative AI has moved from being an experimental idea to a proven, evidence-backed solution, with recent studies demonstrating its tangible value in practice [53]. This transition highlights its role not only in addressing data limitations but also in supporting more reliable and adaptive decision-making across complex industrial contexts.

3. Review scope and methodology

A formal systematic literature review was not feasible for examining the nexus of AI realization, performance, and the O&G industry, as no prior study has directly addressed this specific combination, with the partial exception of [13]. In addition, references were not restricted to academic publications. Accordingly, this article adopts a thematic literature review, which, unlike a systematic review [54], does not apply formal risk-of-bias assessments and instead emphasizes synthesizing fragmented evidence into a structured framework. The review has two objectives: to consolidate existing research on AI realization and its performance impacts, and to map these insights onto the distinctive context of the O&G sector.

Relevant literature to the targeted themes was identified through searches in *Scopus* and *Web of Science*, covering publications from 2010 to 2024, a period selected because AI began to mature around 2010

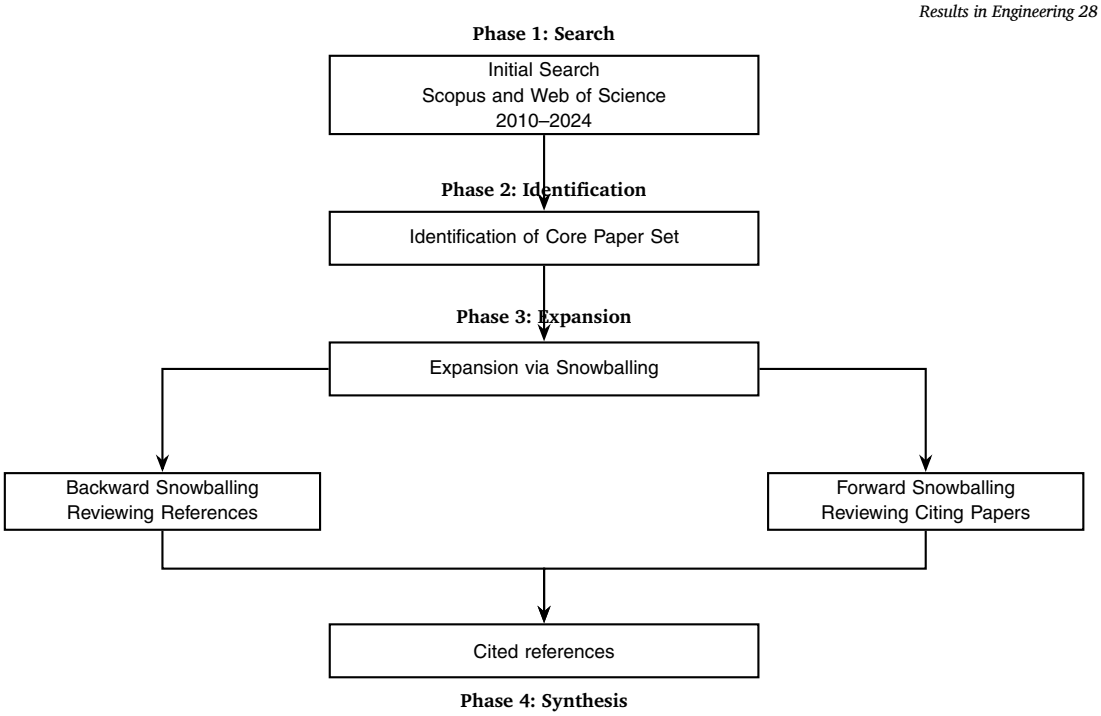


Fig. 3. Iterative literature search and selection process using the snowballing method.

Table 3
Core papers identified for the three thematic strands of the review.

Theme	Core papers and rationale
Drivers of AI Realization	Mikalef and Gupta [10] – Seminal on capabilities Koroteev and Tekic [56] – O&G-specific drivers Waqar et al. [2] – Recent, broad O&G AI review
Measuring AI Realization	Mikalef and Gupta [10] – Capability index Hussain et al. [31] – Textual analytics in energy Chu et al. [57] – Proxy measures (patents)
Performance Outcomes	Lu et al. [1] – O&G performance Chen et al. [58] – Environmental outcomes Bharadwaj et al. [59] – Seminal on IT & financial performance

[55]. The search strategy combined keywords and Boolean operators associated with the three reviewed themes:

1. **AI realization**: ("artificial intelligence" OR "machine learning" OR "deep learning") AND (workforce OR skills OR management OR strategy OR governance OR "organizational change"),
2. **Measuring AI realization**: ("artificial intelligence" OR "machine learning" OR "deep learning") AND ("AI measure" OR "AI gauge"),
3. **Performance outcomes**: ("artificial intelligence" OR "machine learning" OR "deep learning") AND ("performance" OR "efficiency" OR "productivity" OR "sustainability").

From this process, a set of core papers most relevant to the above three themes was identified. These studies served as the foundation for the review. A snowballing approach was then applied. Both backward snowballing (reviewing the references cited in the core papers) and forward snowballing (reviewing studies that cited the core papers) were applied to expand coverage and capture additional relevant contributions. Snowballing also enabled the inclusion of sources that would not have been accessible through academic databases alone, such as company websites, annual reports, and industry white papers. This process specifically gathered evidence from these non-academic sources to support and contextualize our thematic analysis. This iterative process facilitated the inclusion of both foundational and recent influential works. Fig. 3 summarizes the overall search and selection process, while Table 3 presents the core papers that anchored each theme.

4. Drivers to AI realization in the O&G industry

Adoption of AI tools alone does not create value. Progress toward AI realization depends on three interconnected groups of drivers that shape how quickly businesses develop the necessary data, skills, and routines. These drivers are related to people, technical readiness, and managerial orientation, each playing a distinct yet interconnected role in AI outcomes.

4.1. People

People are a key driver of AI realization in the O&G sector. Without the right skills, acceptance, and governance, even advanced technical systems may fail to deliver meaningful value. Three main people-related factors shape the pace and quality of AI adoption and integration:

1. Specialist talent. Upstream seismic analysis, midstream integrity analytics and downstream process control all need hybrid domain–data skills that remain scarce and costly [6,60,61]. Data science

courses were not part of the academic curriculum; therefore, 1) academic curriculum should accommodate AI courses, and 2) companies should train their engineers to use AI skills [62].
2. Workforce acceptance. Field crews may see AI as a threat unless benefits and safety gains are made clear [56,6]. Clear training and co-design help build trust and reduce resistance.
3. Ethical and legal readiness. AI advice that misfires can raise liability questions for operators, vendors and regulators. Early work on "responsible AI" stresses the need for explicit guidelines on transparency and accountability [63].

4.2. Technical

Technical readiness is a critical enabler of AI success in O&G. Even with strong leadership and skilled people, it is difficult to achieve meaningful results without reliable data, adequate infrastructure, and robust algorithms. Three technical factors are especially important:

1. Data quality and integration. Sensor readings, well logs and trading feeds are often siloed and inconsistently labelled [64,65]. Cleansing and joining these sources are the first hurdles to useful models.
2. Infrastructure. High-bandwidth links to rigs, edge devices and cloud platforms enable large-scale training and fast inference [1]. Firms must achieve a high level of machine learning operations (MLOps), which is the set of practices that manage the full life-cycle of machine-learning models—from data preparation and model development through deployment, monitoring, and continual improvement [66]. Firms without mature MLOps pipelines struggle to keep models current.
3. Robust algorithms. Geological variability and sparse failure data can cause over-fitting; physics-guided networks, transfer learning and continual updates are therefore essential [67,33]. Community playbooks and shared benchmarks shorten the learning curve [68].

4.3. Managerial

Managerial orientation plays a key role in translating AI potential into real outcomes. Clear strategy, effective collaboration, and responsiveness to external pressures help steer AI projects beyond isolated trials. Three managerial aspects often shape progress:

1. Strategic vision. AI projects that are not clearly tied to business goals stay as small, isolated efforts [69,70]. Strong leadership support and funding rules that encourage reusing solutions and sharing data rather than building single-use tools will help these efforts grow into company-wide solutions.
2. Cross-functional collaboration. Geologists, process engineers, and IT teams must share data and jointly own models; entrenched silos slow diffusion [70]. Companies that shift to agile, product-team structures report faster pay-offs [51].
3. External pressure. Safety rules, carbon targets, and price swings all push managers to look for new efficiencies [2]. Firms facing strict regulation often invest earlier in monitoring and decision support, while those in stable markets can lag behind [32]. Open partnerships with suppliers, universities, and peers spread cost and risk, echoing long-standing innovation findings [71].

5. Measuring AI realization

A company's level of AI integration into its daily operations cannot be determined by a single data stream. Therefore, three measurement families have been used in previous research to assess this level, as shown in Fig. 4. These measures are introduced first, followed by a discussion about the relation between AI drivers and these measures.

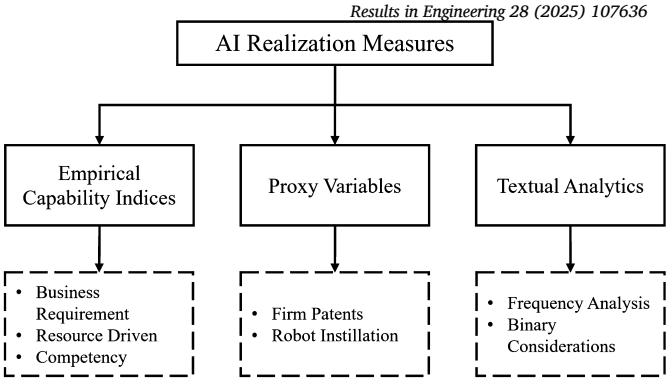


Fig. 4. Main measurement families for firm-level AI realization.

5.1. Textual analytics

These methods rest on the availability of full-text reports that reflect an organization's capabilities and established natural language processing (NLP) pipelines [72–74]. The U.S. SEC Form 10-K reports are an example of such reports. These reports produce two indicators:

1. Frequency score. A count (often normalised) of AI-related words; higher counts imply stronger strategic emphasis [75–77].
2. Adoption flag (AIA): A binary variable that turns to 1 in the first year upon which AI is mentioned in company disclosures and remains 1 thereafter [31].

5.2. Proxy variables

1. AI patent counts. Researchers count the number of AI-related patents a company files each year by matching keyword lists to its patent records. This serves as a proxy for the company's AI-related R&D activity and accordingly AI realization. [57,78].
2. Robot density. Vendor or agency statistics on robot installations/imports are scaled by industry employment to approximate embodied automation and the progression of AI realization over time [58,79–81].

5.3. Empirial capability indices

Grounded in Resource-Based Theory (RBT), competitive advantage rests on how well a firm assembles and leverages valuable, rare, and hard-to-imitate resources. Applying this logic to artificial intelligence, researchers view AI capability as the composite capacity to turn data, algorithms, people, and processes into operational value. Schmidt [17] describes it as the skill to combine "data, techniques, individuals, and procedures" so that AI tools are deployed and used effectively, while Mikalef [10] emphasises an organisation's proficiency in exploiting all AI-linked assets to improve performance. The concept, therefore, widens the lens from technology alone to the full bundle of organisational resources required to unlock AI's potential. Researchers have considered three types of indices to measure AI realization: 1) resource-driven capabilities, 2) business requirement capabilities, and 3) AI competency.

5.3.1. Resource driven capability

A firm's level of AI realization is proxied by assessing resources based on the RBT model, as shown in Fig. 5. Resources are classified into three types: 1) tangible, e.g., infrastructure and data, 2) human, e.g., skills, and 3) intangible, e.g., coordination and learning culture [10,82,83].

Tangible resources, e.g., data assets, hardware, software, and platform infrastructure, are usually gauged with straightforward, count-based metrics (e.g., terabytes stored, number of servers, cloud-service spend). Although these measures capture a firm's basic digital and technological capacity, such assets are widely purchasable on the open mar-

Table 4
Constructs of AI as a Resource Driven Capability [10].

Construct	Definition
Data	Measure an organization's availability to data at the appropriate level of granularity and its ability to integrate and efficiently cleanse data for AI applications.
Technological	Denotes whether a company has invested in the requisite hardware and software to provide flexible data storage, analysis and processing internally and externally.
Technical & Business Skills	Assesses extent of AI-specific competencies among technical and business personnel.
Inter-Departmental Coordination	Assesses the level of open communication and collaboration across departments, as well as the degree of their shared vision.
Organizational Change	Denotes the extent to which an organization can swiftly adjust to changing circumstances, whereas risk proclivity assesses the company's inclination to undertake high-risk initiatives that may result in substantial returns.

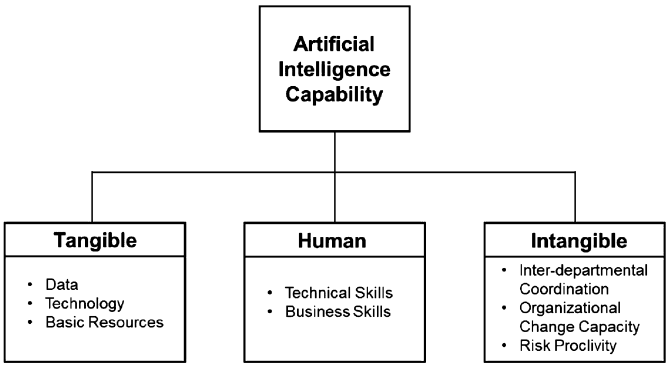


Fig. 5. Artificial intelligence capability based on the RBT model, adapted from [10].

ket and, by themselves, seldom confer lasting competitive advantage [10].

Organisations typically gauge their human capital by reviewing employees' knowledge, skills, experience, leadership, vision, communication and cooperation abilities, and problem-solving capacity [10]. Digital-capability studies highlight technical and business skills as the key parts of this human-resource base [59,84]. When assessing AI capability, the main human indicators become the number and depth of staff with AI-specific technical and commercial skills.

Intangible resources are believed to be more challenging to duplicate and of greater significance in uncertain and turbulent markets [85]. In contrast to the other two resource categories, intangibles are far more elusive and challenging to locate within firms [86]. Organizations cultivate intangible resources by a distinctive amalgamation of organizational history, personnel and procedures, which enhances their heterogeneity and non-replicability [10].

Table 4 describes measurable constructs stemming from the literature, reports and interviews [82]. A survey instruments were developed to quantify these constructs [10].

5.3.2. Business requirement capabilities

Employing the same RBT perspective, three AI capabilities can be evaluated as indicators of AI relization by a firm, corresponding to three essential business requirements: automating operations (automation), acquiring insights (analytics), and interacting with consumers (relational) [86]. These capabilities present distinct advantages as AI-driven automation is proficient in standardisation by automating repetitive and routine tasks; AI-driven analytics is adept at personalisation, identifying patterns and interpreting their implications in decision-making; and AI-driven relational capability is optimal for cultivating robust relationships, learning, and adapting its behaviours through interactions with human users [77,87].

Table 5
Relative influence of driver groups on AI realization techniques.

Metric family	People	Technical	Managerial
Textual analytics	low	indirect	high (tone, disclosure)
Patents / robots	medium	high	medium
Capability indices	high	high	high

5.3.3. AI competency

AI competency was defined as the firm's capacity to effectively integrate its AI technology, skills, knowledge, and other complementary resources, potentially establishing a distinctive advantage over rivals [83]. It can be measured using three dimensions: infrastructure, business spanning, and proactive stance. The infrastructure dimension encompasses a firm's capacity to securely manage its data assets and the essential infrastructure required to transform raw data into valuable AI applications. The business spanning dimension denotes management's ability to leverage AI to advance business objectives from conception to execution. The proactive stance revolves around the capacity to utilize AI and the ongoing pursuit of methods to achieve the same.

5.4. Drivers and AI realization measures

People skills, technical setup, and management approach affect each type of metric differently, as shown in Table 5.

- Textual scores are highly exposed to managerial "impression management" and disclosure tone; they could signal intent more than verified capability [72,88].
- Patent and robot proxies emphasise engineering/R&D effort and automation spend, potentially overweighting firms that internalise innovation [78,58].
- Capability surveys cover the resource bundle but risk optimism and common-method bias when responses come mainly from managers [89,90,10].

5.5. Triangulation rule for AI realization measures

Text-based signals facilitate longitudinal tracking, documenting the progression of organizational discourse and strategic focus on AI over time. Proxy counts could potentially serve as benchmarks, facilitating comparability across various organizational contexts. Capability surveys provide flexibility for cross-sectional and cross-geographical analyses, capturing subjective perceptions cross various contexts. Thus, no one measure can capture the full picture about AI realization in firms; moreover, any single indicator can be biased by people, technical, or managerial factors. Therefore, researchers should triangulate across measurement families (e.g., combine text signals, proxy counts, and capability surveys) to strengthen validity and limit common-method bias [91,92,89].

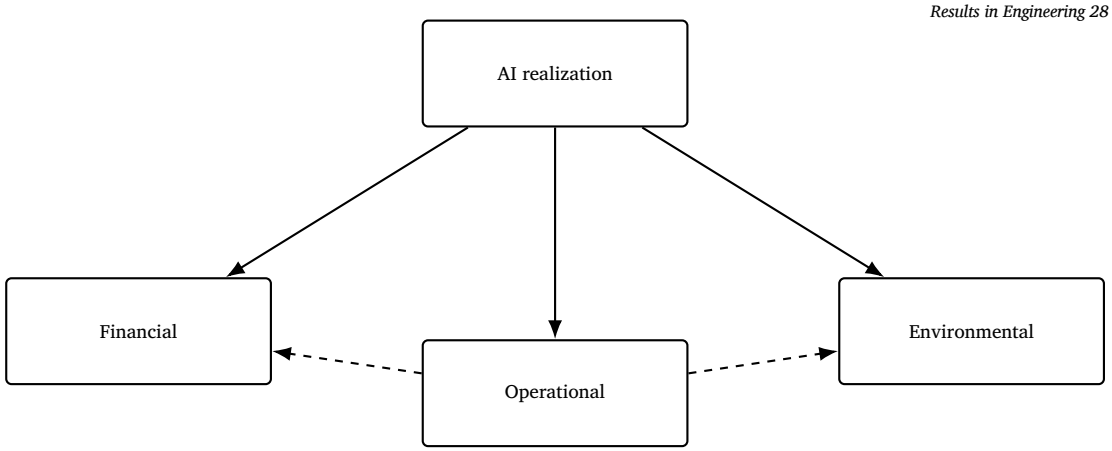


Fig. 6. Impact of AI Realization on company performance.

6. Performance outcome domains

Only when AI realization yields observable outcomes does it matter. Prior research has grouped these outcomes into three domains that are interlinked. Below is a brief summary of each domain, including key metrics and the primary cause-and-effect reasoning. Fig. 6 illustrates how improvements in operational performance frequently cascade down to financial and environmental ones [93].

The following summary describes each performance measure:

1. Operational. Compared to financial accounts, efficiency measures are more accessible to the work floor and frequently show advantages sooner. Delivery lead time, equipment uptime, and staff engagement are examples of common indicators [94,95]. In the O&G industry, AI-guided monitoring has reduced unproductive time in upstream drilling, while predictive maintenance increases refinery throughput downstream [1].
2. Financial. Work tracks both internal accounting ratios, e.g. Return on Assets (ROA) and Return on Equity (ROE), and market measures such as abnormal stock returns [59,96,97]. Although early-stage capability build-up can lower short-run profits, AI is expected to increase revenue through improved forecasts and reduced expenses.
3. Environmental. Rising computing demand makes it essential to track a firm's energy use and emissions. Common indicators are energy intensity (output per kWh), firm- or city-level CO_2 and SO_2 releases, and the share of green patents in the company's total portfolio [58]. Business intent can also be read from sustainability reports by counting words and phrases linked to the energy transition [15].

6.1. Triangulation rule for performance measures

Credible evaluation of AI performance domain should comprehensively analyse operational, financial, and environmental performance outcomes altogether as improvements in operational efficiency can enhance profitability; however, they may also elevate data processing requirements and energy consumption, which could adversely affect environmental performance. Financial gains, including enhanced ROI, may rely on continuous operational innovation and customer satisfaction. Environmental benefits, such as reduced emissions, are significant, primarily when they coincide with operational efficiency and cost-effective implementation. Isolating these aspects would obscure their interdependencies and result in an incomplete understanding of AI's overall value.

7. Conceptual synthesis

The elements of a conceptual synthesis for analysing how AI is implemented and what it delivers in the O&G industry are shown in Fig. 7. To realise AI effectively, three driver sets—people skills, data/technological preparedness, and managerial vision—are essential. When these capabilities are in place, AI value is expected to surface in three areas. The first visible gains are usually operational (e.g., faster cycle times, fewer unplanned shutdowns). These gains support financial outcomes (higher ROA, stronger cash flow) and can also lower environmental impacts (reduced energy per barrel produced, lower CO_2 emissions).

Inputs to this conceptual synthesis are the three driver groups; outputs are the three performance domains. Researchers can gauge AI realization with capability-survey scores, patent or robot counts, and the company's disclosures keyword trends as intent signals.

Although the above synthesis applies broadly, the O&G industry faces specific conditions that impact both drivers and realised effects, as marked in Fig. 7:

- Ownership form. IOCs have the advantage of worldwide integration and lack of constraints from bureaucratic procedures and political factors compared to NOCs, which may face policy ceilings or lengthy approval chains [31,98].
- Value–chain segment. O&G segments have unique AI objectives and performance measures. Upstream focuses on exploration optimization, midstream on logistics and risk, and downstream on refining and customer innovation [56,35].

To enable effective AI realization, decision-makers can target investments in people, data/technology, and change management according to business type and value-chain position. Operational improvements should be tracked alongside financial and environmental results to judge whether AI investments are paying off. When executives report AI plans in public filings, accuracy matters because such disclosures are used as one metric of AI realization. Proxy indicators can provide feedback on AI realization, but capability indices can be argued as a closer reflection of AI resources' maturity.

7.1. Recommendations for O&G companies

In addition to the recommendations about how to manage the three drivers in Section 4, decision-makers in O&G companies, need also to do the following:

1. Link the data pipes. Integrate operations, finance, and sustainability databases so all teams work with one common view [1,65], which enhances collaboration, sparks innovation, lowers costs, and improves operational transparency.
2. Run annual capability scans. A short survey on skills, data quality, infrastructure, and governance provides an "AI-readiness" scorecard; tracking it over time highlights gaps early [10].

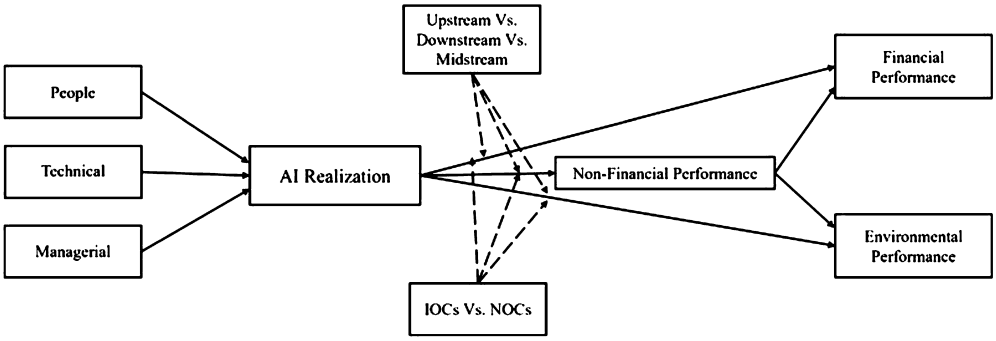


Fig. 7. A conceptual thesis of the three components responsible for understanding how AI would affect the organizational performance of companies in the O&G sector.

7.2. Recommendations for the O&G industry

O&G professional organisations such as the Society of Petroleum Engineers (SPE) can accelerate AI realization across the industry. They should:

1. Compare ownership and field life-cycle. Test whether IOCs, NOCs, green-field assets, or mature fields achieve different AI pay-offs [31,98].
2. Share and replicate. Create open consortia to swap primary, ancillary, and image data, and encourage firms to release code where regulations allow, speeding peer review and adoption [56,68].
3. Grow AI skills. Add data-science modules to engineering and supply-chain courses and study how training hours correlate with firm capability scores [61,60].
4. Set clear reporting rules. Standardise when and how AI initiatives are disclosed; clearer rules reduce hype and enable fair comparison [72,73].

Collectively, these actions will strengthen the evidence base, close capability gaps, and aid in converting AI potential into lasting value for the oil-and-gas sector.

8. Conclusion and future research

Firms in O&G and other industries invest in AI only when it delivers clear and lasting performance gains. These gains require full AI realization—embedding data, models, and workflows across the business—rather than scattered pilot projects. Testing whether AI realization drives better results requires solid metrics on both sides: the extent of realization and the resulting performance shifts. This paper reviewed existing measures of realization, outlined their effects, and highlighted the key drivers that enable organisation-wide AI value. Firms in O&G, or any other industry, invest in AI only if it can lift performance in a clear and lasting way. Such gains depend on full AI realization—embedding data, models, and workflows across the business—rather than scattered one-off projects. To test whether realization truly drives better results, solid metrics are needed on both sides: how far AI is embedded and how performance shifts. This paper reviews the main methods for gauging AI realization and its effects, and it outlines the key drivers that help firms achieve genuine, organisation-wide AI value.

Research to date highlights three layers in the AI-performance link. First, three broad drivers—people skills, technical readiness, and managerial backing—determine how far a firm moves from isolated pilots to full AI realization. Second, progress is typically measured through textual analytics of corporate filings, proxy indicators such as AI-patent or robot counts, and survey-based capability indices. Third, the effects of realised AI show up across operational, financial, and environmental metrics. In the O&G sector, these drivers, gauges, and outcomes are further conditioned by two industry specifics: the position of the activity in the value chain (upstream, midstream, or downstream) and whether the firm is a state-owned NOC or a commercially driven IOC.

To raise AI realization from scattered pilots to firm-wide practice, decision-makers in O&G must act on all three driver groups. For the people driver, engineers need hands-on AI training, clear proof that the tools help rather than replace them, and firm rules for responsible, ethical use. On the technical side, teams must clean and link data silos, build reliable MLOps pipelines, and share datasets and models across assets. Managerial action begins with leaders who champion data sharing, break functional walls, and form partnerships with suppliers and universities. Because no single indicator tells the whole story, progress and impact should be measured with triangulation—combining at least two different gauges of AI realization and matching them to multiple performance metrics.

Because triangulation requires consistent data, firms should first link their operations, finance, and sustainability systems. Next, they can create a short ‘AI-readiness’ scorecard covering skills, data quality, infrastructure, and governance, and update it annually to track progress. Lasting progress also depends on industry-wide cooperation. Professional bodies such as the Society of Petroleum Engineers can compare NOCs and IOCs, promote secure data sharing, encourage universities to add applied-AI courses, and publish clearer reporting rules. These steps give decision-makers a stronger basis for judging—and improving—the real impact of AI. Because triangulation depends on consistent data, firms should first link their operations, finance, and sustainability systems so that every team works from the same facts. Next, they can build a short “AI-readiness” scorecard that rates skills, data quality, infrastructure, and governance, then update it each year to spot gaps early. Lasting progress also requires industry-wide cooperation. Professional bodies such as the Society of Petroleum Engineers can compare results for international versus national oil companies, promote secure data-sharing schemes, press universities to add applied-AI courses, and publish clear guidelines on what AI information companies should disclose. Together, these steps give decision-makers a firmer base for judging—and improving—the real impact of AI.

This paper establishes the basis for empirical testing of AI realization impact on the O&G sector. Future research may utilize a mixed-method design, employing PLS-SEM or hierarchical regression modelling to evaluate causal pathways. AI enablers, such as data infrastructure, skills, and strategic alignment, can be considered independent variables, whereas realized AI capabilities act as mediators. Firm performance outcomes, including financial metrics such as ROA, operational measures like throughput efficiency, and environmental indicators such as CO_2 intensity, could serve as dependent variables. Contextual factors such as ownership structure and regulatory intensity may act as moderators, influencing the strength or direction of relationships. Constructs can be assessed through structured surveys employing validated scales (e.g., AI Capability Index) and validated by secondary performance data and corporate disclosures. Qualitative interviews with executives may yield further insights into organizational dynamics and decision-making, thereby

enriching the interpretation of statistical results and enhancing the practical relevance of the conceptual synthesis.

CRediT authorship contribution statement

Mohamed Abdalla AlAbdouli: Writing – original draft, Methodology, Conceptualization. **Sameh Al-Shihabi:** Writing – review & editing, Supervision, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used *Grammarly* for proofreading and *ChatGPT* (OpenAI) to improve readability and clarity. After using these tools, the authors reviewed and edited the content as needed, and take full responsibility for the final content of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No new data were created or analyzed in this study. Data supporting the findings of this study are available from the corresponding author upon reasonable request.

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